

tion. The power train used for the MCP1630 is a SEPIC. The module can take inputs anywhere from 10V to 28V and up to 16 modules can be daisy-chained for charging additional cells.

The MCP1630 multi-bay Li-ion charger has the flexibility to optimise the charging algorithm for new battery technology and add proprietary features by coding the microcontroller. The charger offers multiple charging features, such as input overvoltage protection, battery overvoltage protection, thermal shutdown, battery-temperature monitoring, and battery-pack fault monitoring. It can also detect the status of the battery along with the insertion and removal of the battery pack.

The brain of the charger is PIC18F2410, which is an enhanced flash microcontroller with 10-bit A/D. The microcontroller runs at a frequency of 40MHz and provides high computational performance along with lower power consumption.

The reference design also has programming header pins required for updating the firmware of the microcontroller. It also employs a dual op-amp MCP6292. The battery charger automatically detects the insertion or removal of a battery pack.

Important components.

U1, U4 - MCP1630 charging IC
 U2, U5 - MCP6292 dual op-amp MSOP8
 U3, U6 - SN74LVC1G66DBVR bilateral switch
 U7 - NC7SZ74K8K D-type flip-flop
 U8 - PIC18F2410 8-bit CMOS microcontroller
 U9 - SN74LVC2G14DBVR dual inverter

Features of MCP1630.

- Programmable parameters modified in firmware
- The MCP1630 multi-bay Li-ion charger has the flexibility to optimise the charging algorithm for new battery technologies and add proprietary

features by coding the microcontroller

- Controlled soft-start

Safety features.

- Overvoltage protection (battery removed)
- Overcharge protection to prevent the battery from becoming dangerously overcharged
- Overcurrent protection in the event of a shorted battery
- Battery reversal protection
- Input short-circuit protection
- Overtemperature protection to prevent the battery from reaching too high a temperature during charge
- Fast charge termination if the battery temperature is outside a preset window
- Controlled soft-start

The module is suitable for charging a Li-Ion battery pack. You can find more details and design resources by scanning the QR code.



EV2639A-R-00B 2S cell charger with integrated boost converter

The EV2639A-R-00B from Monolith Power is the reference design board for the MP2639A, which is a 2-cell battery charge management system. MP2639A is a highly integrated, flexible switch-mode charge management IC that can take input between 4V and 5.75V and charge 2S Li-ion and Li-Po cells. The module is capable of charging 2S cell with 2.5A charge current.

The reference module works in a charge mode and based on the state of the input it determines the management of the system and battery charging. The module also provides voltage based fuel gauge indication through four LED drivers present in the circuit.

The MP2639A IC is highly programmable, whose input voltage

and current can be controlled by updating the firmware. It gives an option of enhancing the discharge current up to 5A. The board offers multiple safety features, which include input overvoltage protection, battery overvoltage protection, thermal shutdown, battery temperature monitoring, and a programmable timer to prevent prolonged charging of a dead battery.

In this reference design, the MP2639A works in a step-up mode to charge the 2-cell battery from the 5V input. In absence of the charger, the module is in discharge mode, where it works in a step-down mode to give an output of 5V. For the charging function, the battery charging IC detects the battery voltage and based on the voltage charges the battery in three phases: trickle current, constant current, and constant voltage.

Features.

- Bidirectional operation for charging or discharging
- 2-cell Li-ion or Li-polymer charging
- Cell balance and protection escorting for two-series cell charging
- Light-load detection for discharge mode
- Integrated voltage based fuel gauge using four LED drivers
- Takes 5V input and charges 2S cell

Safety features.

- Input overvoltage protection
- Battery overvoltage protection
- Thermal shutdown
- Battery temperature monitoring
- Programmable timer to prevent prolonged charging of a dead battery

This charger reference design can be used in various devices, including mobile, tablets, power stations, mobile internet devices, and standalone Li-ion battery chargers. You can find additional details and

